Level 2



CLOUDFLIGHT CODING CONTEST

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LEVEL 2

THE PROFESSOR NEVER CHECKS THE SHELVES

Why look when you can just react? Whenever a reaction needs something that the lab doesn't have, a pneumatic tube in the ceiling goes THWUMP and the supply shop delivers it on the spot. The bill goes straight onto the Professor's tab.

This month's invoice is suspiciously large, and nobody remembers what's still sitting on the shelves.

With the professor’s notebook in your left hand, the shop’s catalogue in the other, you set out to investigate.

Find out what it all cost, and what's left over, before the accountant does.

- You are given a list of molecules that can be bought at a stated price
- You are given a list of reaction sequences, for each:
  - Start with an empty inventory
  - Perform the reactions one by one
  - Reactions consume input molecules if they are in the inventory, otherwise they are bought automatically
    - *All given reactions will only use molecules that are present in the inventory at the time they are performed or that are buyable*
  - **Determine the total cost of the bought molecules and the final state of the inventory**



Reaction example

Cost	Molecule
5	H2
1	O2
3	CO2
15	H2O

Reaction	Inventory	Total Cost	Cost breakdown
-	-	0	
2 H2, 1 O2 -> 2 H2O	2 H2O	11	2*5 from H2 + 1 from O2
2 H2, 1 O2 -> 2 H2O	4 H2O	22	11 + 2*5 from H2 + 1 from O2
6 CO2, 6 H2O -> 1 C6H12O6, 6 O2	1 C6H12O6, 6 O2	70	22 + 6*3 from CO2 + 2*15 from H2O
2 H2, 1 O2 -> 2 H2O	1 C6H12O6, 5 O2, 2 H2O	80	70 + 2*5 from H2

Input

Name		Description	Example
M		Number of buyable molecules	4 5 -> H2 1 -> O2 3 -> CO2 15 -> H2O 3 1 2 H2, 1 O2 -> 2 H2O 4 2 H2, 1 O2 -> 2 H2O 2 H2, 1 O2 -> 2 H2O 6 CO2, 6 H2O -> 1 C6H12O6, 6 O2 2 H2, 1 O2 -> 2 H2O 4 2 H2, 1 O2 -> 2 H2O 2 H2, 1 O2 -> 2 H2O 2 H2, 1 O2 -> 2 H2O 6 CO2, 6 H2O -> 1 C6H12O6, 6 O2
Cost to molecule (Repeated M times)		The cost and a molecule separated by an arrow “ -> ”	
Empty line			
N		Number of cases	
repeated N times	R	The number of reactions	
	Reaction (Repeated R times)	Two molecule list separated by an arrow “ -> ”	
	Empty line between cases		

Output

Name	Description	Example
Cost to resulting inventory (repeated N times)	The total cost of all reactions and the resulting inventory separated by an arrow “ -> ”	11 -> 2 H2O 80 -> 5 O2, 2 H2O, 1 C6H12O6 51 -> 6 O2, 1 C6H12O6



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